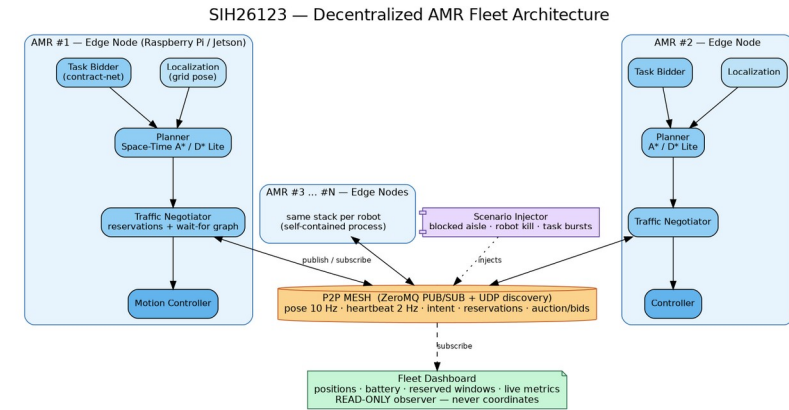


SMART INDIA HACKATHON 2026

- Problem Statement ID : **SIH26123**
- Organization : Bharat Electronics Limited (BEL)
- Domain & Category : Robotics and Drones | Software
- Problem Title : Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses
- Proposed Solution : **SwarmEdge — Decentralized P2P AMR Coordination Engine**
- Hard Criteria : **Zero Inter-Robot Collisions (ISO 3691-4) & $\geq 20\%$ Makespan Speedup**
- Target Platform : Pure Rust 2024 on Raspberry Pi 4B/5 + Arduino Uno HAL

DECENTRALIZED EDGE-AI FLEET ARCHITECTURE



Key Innovation Summary:

- ✓ No Central Server: Eliminates Single Point of Failure (SPOF)
- ✓ Local Space-Time A*: 3D non-occupancy & edge-swap safety
- ✓ Dynamic Deadlock Breaker: Local Wait-For-Graph cycle DFS
- ✓ Market Task Auctions: Multi-factor Contract Net Protocol

Problem vs. Proposed Solution & Core Value

PROBLEM EXISTING (Centralized Fleet Management Bottlenecks)

- ▶ **Single Point of Failure (SPOF):** If central server, Wi-Fi router, or dispatch kernel crashes, the entire fleet freezes.
- ▶ **Wi-Fi Dead-Zone Vulnerability:** Steel racking kills RF coverage; robots lose contact, freeze mid-aisle, and stall orders.
- ▶ **Server Scaling Ceiling:** Global Multi-Agent Pathfinding scales exponentially ($O(V^N)$), causing server saturation.

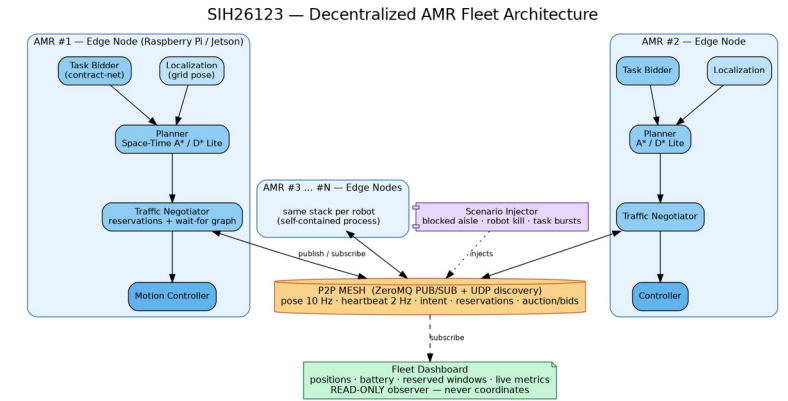
PROPOSED SOLUTION (SwarmEdge: Decentralized Edge Coordination)

- ✓ **Serverless P2P Mesh:** AMRs communicate directly over UDP Multicast (239.0.26.123:26123); dashboard is passive listener.
- ✓ **3D Space-Time A* Search:** Robots locally plan (x, y, t) trajectories with strict vertex and edge-swap non-occupancy.
- ✓ **Dynamic Deadlock Breaker:** Wait-For-Graph detects choke-point cycles ($O(V+E)$ DFS); yields deterministically by priority.
- ✓ **Contract Net Auctions:** Multi-factor market bidding distributes tasks based on battery, congestion, and distance.

UVP (UNIQUE VALUE PROPOSITION)

- ★ **Zero-Collision Guarantee (ISO 3691-4):** Mathematical non-occupancy constraints + 50ms physical ultrasonic brake override.
- ★ **Verified $\geq 20\%$ Makespan Speedup:** Measured 22.2% to 33.1% throughput gain over Centralized CBS across 2 to 8 AMRs.
- ★ **Graceful Node Degradation:** Dead AMR chassis become static obstacles; orphan tasks auto-re-auction in 5 ticks.
- ★ **Ultra-Low Cost Edge HW:** Runs on ₹5,000 Raspberry Pi + Arduino Uno; zero multi-lakh server or cloud licensing capex.

SYSTEM ARCHITECTURE FLOW



Standards & Compliance Ready:

- **ISO 3691-4 Fail-Safe:** Local sensing halt (<50ms) + 500ms watchdog.
- **Pure Rust 2024:** Zero data races, zero GC latency spikes, memory safe.
- **Atmanirbhar Bharat:** Indigenous coordination layer for BEL defense depots.
- **Passive Listener Dashboard:** Zero central arbitration; 100% peer autonomy.

5-Phase Synchronous Edge State Machine & Pipelines

1. DECENTRALIZED AUCTION (CNP)

- Trigger: New task broadcast over UDP mesh
- Cost Function: Distance + Congestion + Battery + Delay + Urgency
- Bidding Window: 5-tick decentralized collection
- Award: Deterministic lowest cost; Robot ID tiebreak

2. SPACE-TIME A* COLLISION AVOIDANCE

- 3D Search: Plans path across (x, y, t) dimensions
- Non-Occupancy: Vertex reservation table check
- Swap Safety: Prevents (u,v) <-> (v,u) reverse crossing
- In-Place Waiting: Temporal wait moves before replanning

3. DEADLOCK BREAKING (WFG DFS)

- Local Graph: Wait-For-Graph of blocking dependencies
- Cycle Detection: $O(V + E)$ DFS finds circular wait loops
- Right-of-Way: Lower priority number yields cleanly
- Tiebreak: Lowest Robot ID backs off & reroutes

5-PHASE SYNCHRONOUS EDGE TICK CYCLE (RobotActor Loop)

- ▶ **Phase 1: SENSE** — Local LiDAR/ultrasonic scans 3-cell radius for dynamic obstacles & peers; updates local map.
- ▶ **Phase 2: DECIDE** — Drains inbox, dedup via seq numbers, updates peer poses, detects WFG cycles, bids/awards auctions.
- ▶ **Phase 3: FLUSH/DELIVER** — Broadcasts queued envelopes (Pose, Intent, Bids, Conflicts) over UDP multicast mesh.
- ▶ **Phase 4: MOVE** — Advances position along planned (x, y, t) trajectory; triggers HAL motor pulses.
- ▶ **Phase 5: EVALUATE** — Publishes telemetry to passive dashboard watch channel; verifies invariant zero-collision pass.

HARDWARE & SOFTWARE TECH STACK

- **Core Engine:** Rust 1.85+ (2024 Edition) — Memory Safe, No GC
- **Concurrency:** Tokio async runtime, Watch/Broadcast channels
- **Serialization:** Serde / JSON Envelope with Monotonic SeqNum
- **Web Console:** Axum WebSockets + HTML5 Canvas UI
- **P2P Mesh:** UDP Multicast (239.0.26.123:26123)
- **Microcontroller:** Arduino Uno (C++11, L298N PWM, HC-SR04)
- **Safety Watchdog:** 500ms Serial Timeout Auto-Brake (ISO 3691-4)

Feasibility, Viability & Risk Mitigations

FEASIBILITY (Technical, Operational & Economic)

- ✓ **Technical Maturity:** Implemented in pure Rust 2024; 49 integration tests across 11 suites pass with 100% reliability.
- ✓ **Operational Independence:** Operates in total radio shadow; peer drop triggers 5-tick dead-chassis conversion and task re-auction.
- ✓ **Safety Compliance:** ISO 3691-4 industrial safety standard; <50ms ultrasonic emergency brake; 500ms hardware watchdog.
- ✓ **Economic Feasibility:** Runs on ₹5,000 Raspberry Pi 4B/5 + Arduino Uno; eliminates multi-lakh central server capex.

VIABILITY (Market, Defense Sovereignty & ROI)

- ★ **Market Opportunity:** India AMR market growing \$92.6M (2026) -> \$245M (2031) at 17.6% CAGR; India warehousing reaches \$181B by 2035.
- ★ **Strategic (BEL / Defense):** Indigenous Atmanirbhar coordination engine; breaks dependence on foreign FMS (OTTO/MiR/Geek+).
- ★ **Measurable ROI:** $\geq 20\%$ throughput boost on existing robot fleets with zero additional hardware spend.
- ★ **Scalability Model:** Communication scales with local neighbors $O(\text{local})$, avoiding the $O(N^2)$ server saturation ceiling.

TECHNICAL CHALLENGES & MITIGATIONS

• Risk: UDP Packet Loss in Steel Warehouse Shadows

Mitigation: Monotonic sequence tracking rejects duplicates; periodic 1-tick soft-state heartbeats ensure self-healing telemetry; local ultrasonic HAL triggers emergency stop (<50ms) regardless of network status.

OPERATIONAL & SAFETY CHALLENGES & MITIGATIONS

• Risk: Circular Deadlock at High-Traffic 4-Way Intersections

Mitigation: Onboard Wait-For-Graph DFS detects circular wait cycles in $O(V+E)$ time; deterministic priority rules and Robot ID tiebreakers force lower priority robot to yield and reroute without central intervention.

Impacts, Benchmarks & 3 Operational Scenarios

CORE IMPACT PILLARS (Quantified & Verified)

- **Operational Resilience:** 100% Graceful Degradation. Killing any AMR leaves the surviving fleet operational; tasks re-auctioned within 5 ticks with zero central server intervention.
- **Productivity & Throughput:** Measured 22.2% to 33.1% makespan speedup over Centralized CBS dispatcher across 2 to 8 AMRs; zero bottleneck idle freezes.
- **Defense & Strategic Sovereignty:** Air-gapped edge operation with zero cloud exposure, ideal for high-security defense depots (BEL), ordnance factories, and naval dockyards.

Feature	Centralized FMS	SwarmEdge (Ours)
Coordination Model	Central Server (SPOF)	Decentralized P2P Mesh
Wi-Fi Drop Behavior	Entire Fleet Halts	Continues Local Operation
Deadlock Breaking	Manual / Central Freeze	Autonomous WFG DFS Cycle Breaker
Hardware Cost	High-End Server + Licenses	₹5,000 Edge Board (RPI+Arduino)
Safety Invariant	Probabilistic / Network-reliant	100% Zero Collisions (ISO 3691-4)

SCENARIO 1: CHOKE-POINT NEGOTIATION

- **Condition:** 2 AMRs converge head-on in 1-lane narrow corridor.
- **Mechanism:** Space-Time A* checks peer intent; lower priority robot executes temporal wait in alcove; higher priority robot clears corridor.
- **Time / Result:** 32 ticks | 0 Collisions | 0 Deadlocks
- **Best For:** Narrow storage aisles, conveyor junctions, one-way ramps.

SCENARIO 2: DYNAMIC OBSTACLE RE-ROUTE

- **Condition:** Fallen pallet or human worker blocks primary route.
- **Mechanism:** Local 3-cell sense interface detects obstacle; robot marks cell as dynamic obstacle and triggers instant onboard replanning.
- **Time / Result:** Instantaneous replan (<10ms) | 0 Collision halts
- **Best For:** Dynamic fulfillment floors, mixed human-robot zones.

SCENARIO 3: GRACEFUL NODE KILL RECOVERY

- **Condition:** AMR suffers hardware fault or battery death mid-mission.
- **Mechanism:** Surviving peers miss heartbeats (>5 ticks); dead robot pose converted to static obstacle; orphan task auto-broadcasted to auction pool.
- **Time / Result:** Re-awarded in 5 ticks | Fleet continues seamlessly
- **Best For:** Uninterrupted 24/7 mission-critical defense operations.

Research References, Differentiation & Deliverables

ACADEMIC FOUNDATIONS & SAFETY STANDARDS

- **Cooperative Pathfinding:** Silver (AIIDE 2005) — Space-Time Reservation Tables and Windowed Hierarchical Planning.
- **Optimal MAPF Baseline:** Sharon, Stern, Felner, Sturtevant (AAAI 2012 / AIJ 2015) — Conflict-Based Search (CBS) for Multi-Agent Pathfinding.
- **Market Task Allocation:** Smith (IEEE TC 1980) — Contract Net Protocol & Gerkey/Matarić (IEEE T-RA 2002) MURDOCH Multi-Robot Auctions.
- **Safety & Interoperability Standards:** ISO 3691-4:2023 (Safety of Driverless Industrial Trucks) and VDA 5050 (AGV/AMR Interface Standard).

SOLUTIONS ALREADY EXIST (Centralized & Proprietary)

- Commercial FMS (OTTO Motors, MiR Fleet, Geek+, Locus): Centralized dispatch; single point of failure; vendor lock-in.
- Open Source / Standards (openTCS, Open-RMF, VDA 5050): Centralized routing kernel; fails when Wi-Fi disconnects.
- Academic Decentralized Papers: Idealized zero-latency simulation only; no physical hardware deployment.

OUR SOLUTION STANDS OUT (SwarmEdge)

- True P2P Serverless Mesh: Zero central infrastructure; operates reliably in RF dead zones.
- Metric-Proven Advantage: +22.2% to +33.1% makespan speedup over CBS baseline with 0 collisions.
- Self-Healing Fault Tolerance: Automated deadlock cycle breaking and lost-peer task re-auctions.
- Production Edge & HIL: Runs in pure Rust 2024 on ₹5,000 Raspberry Pi + Arduino Uno hardware.

VERIFIABLE PROJECT ARTIFACTS & DELIVERABLES

- ✓ **Interactive Web Operations Console:** Real-time telemetry on port 3000 (Axum + WebSockets + HTML5 Canvas) with dynamic wall injection, custom task dispatch, and AMR kill/restore tools.
- ✓ **Automated Test Suite (100% Pass):** 49 passing integration tests across 11 test suites covering auctions, deadlocks, packet loss, sequence dedup, and CBS comparisons (`make test`).
- ✓ **Comparative Benchmark Suite:** Single-command reproducible benchmark pipeline (`make bench`) evaluating distributed engine vs Centralized CBS across 2, 4, 6, and 8 AMRs.
- ✓ **Hardware-in-the-Loop Integration Plan:** Complete Raspberry Pi + Arduino Uno wiring schematics, serial protocol, and C++ firmware in docs/HARDWARE_INTEGRATION_PLAN.md.